

Write_up_Loss_Optimizer

Problem (learning_rate_tuning):

With learning rate $1e1$, the loss decreases steadily but relatively slowly. With $1e2$, it decays much faster and reaches nearly zero within a few iterations, while with $1e3$ it diverges rapidly as the updates overshoot the optimum and the loss grows explosively.

Problem (adamw_accounting):

(a):

B = batch_size

C = context_length

N = num_layers

D = d_model

H = num_heads

V = vocab_size

$F = d_{ff} = 8/3 D$

$$M_{\text{params}} = 4[2VD + N(2D + 12D^2) + D]$$

$$M_{\text{acts}} = 4[N(8BCD + 4BCF + 2BHCC) + BCD + BCV]$$

$$M_{\text{opt}} = 8[2VD + N(2D + 12D^2) + D]$$

$$M_{\text{total}} = M_{\text{params}} + M_{\text{acts}} + M_{\text{grads}} + M_{\text{opt}}$$

$$= 4[N((56/3)BCD + 2BHCC) + BCD + BCV] + 16[2VD + N(2D + 12D^2) + D]$$

(b): For GPT-2 XL, I use $V = 50257$, $C = 1024$, $N = 48$, $D = 1600$, and $H = 25$. Substituting these values into the expression from part (a), the total memory is

$$M_{\text{total}}(B) = 16,150,761,472 B + 26,168,601,600 \text{ bytes.}$$

$$B_{\text{max}} = \text{floor}((80 * 10^9 - 26,168,601,600) / 16,150,761,472) = 3.$$

Therefore, the maximum batch size is 3.

(c): Let (P) be the total number of trainable scalar parameters. AdamW performs a constant number of elementwise operations per parameter. Ignoring $(O(1))$ scalar computations such as the bias-corrected learning rate, the decoupled weight decay costs 2 FLOPs per parameter, the first-moment update costs 3 FLOPs, the second-moment update costs 4 FLOPs, and the final adaptive parameter update costs 5 FLOPs. Therefore, one AdamW optimizer step costs $14P + O(1)$ FLOPs, where $(P = 2VD + N(2D + 12D^2) + D)$ is the total number of model parameters. This does not include the cost of computing the gradients by the forward and backward passes.

(d):

The effective compute throughput is $(0.5 \times 495 \times 10^{12}) = 2.475 \times 10^{14}$

FLOPs/s. If one forward pass for a single 1024-token sequence costs (3.52×10^{12})

FLOPs, then one training step with batch size 1024 costs

$$[3 \times 1024 \times 3.52 \times 10^{12} = 1.081344 \times 10^{16}]$$

FLOPs, where the factor of 3 accounts for one forward pass and a backward pass that costs twice as much as the forward pass. Over 400K steps, the total cost is $(4.325376 \times 10^{21})$ FLOPs. Therefore the training time is

$$[\frac{4.325376 \times 10^{21}}{2.475 \times 10^{14} \times 3600} \approx 4855]$$

hours, or about 202 days.